

PARCEL CREATION

Automated Return Parcel Processing

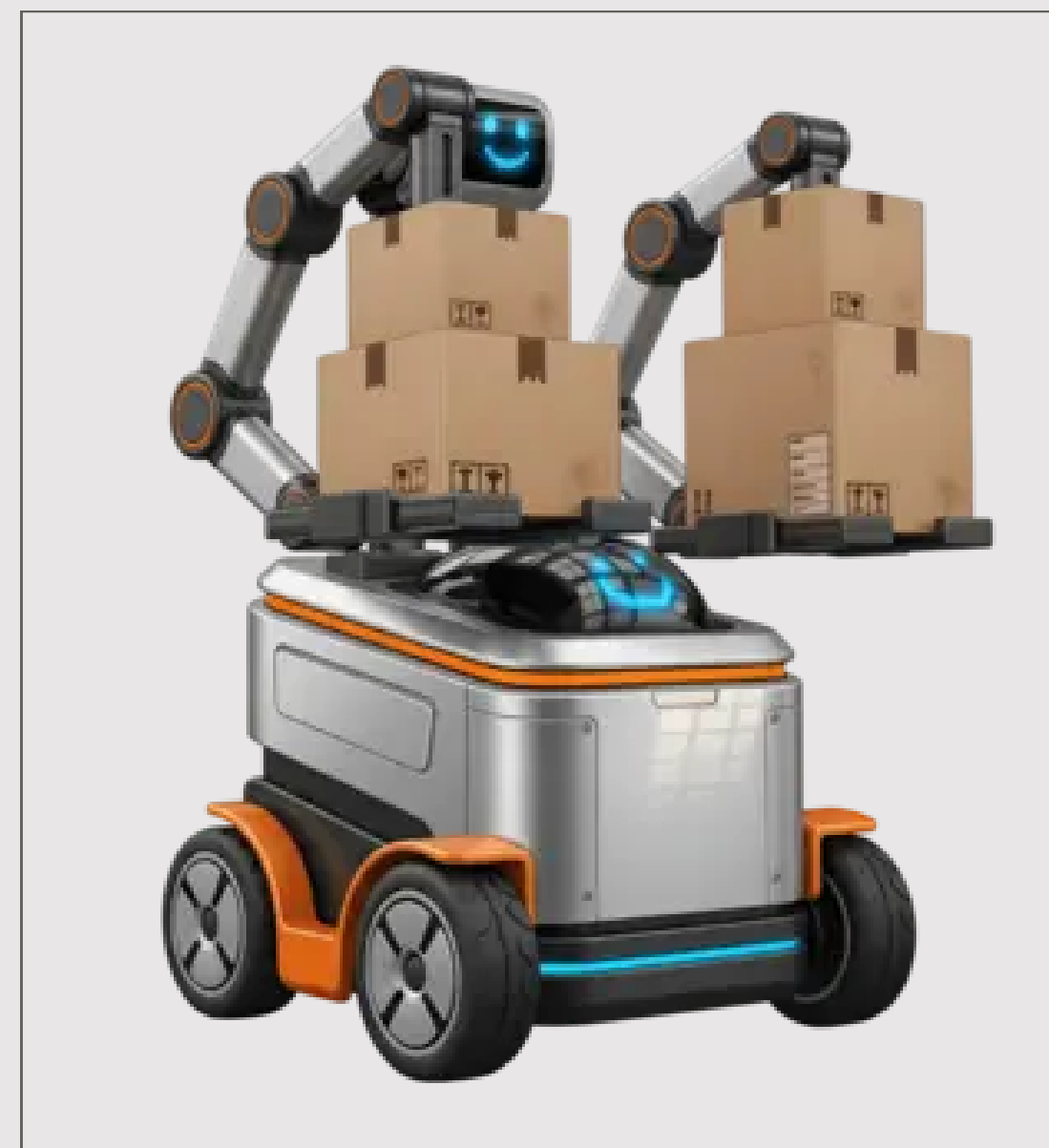
- This project automates end-to-end operational workflows for processing structured data and executing downstream actions.
- The system ingests files, transforms records, interacts with external APIs, and automatically resolves errors using AI-assisted logic.
- The entire pipeline is orchestrated to minimize manual intervention while ensuring reliable, repeatable daily operations.

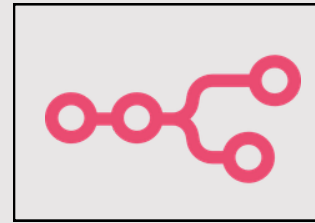
TOOLS

n8n · Azure Blob · PostgreSQL
SendCloud API · Slack · AI
(Gemini)

IMPACT

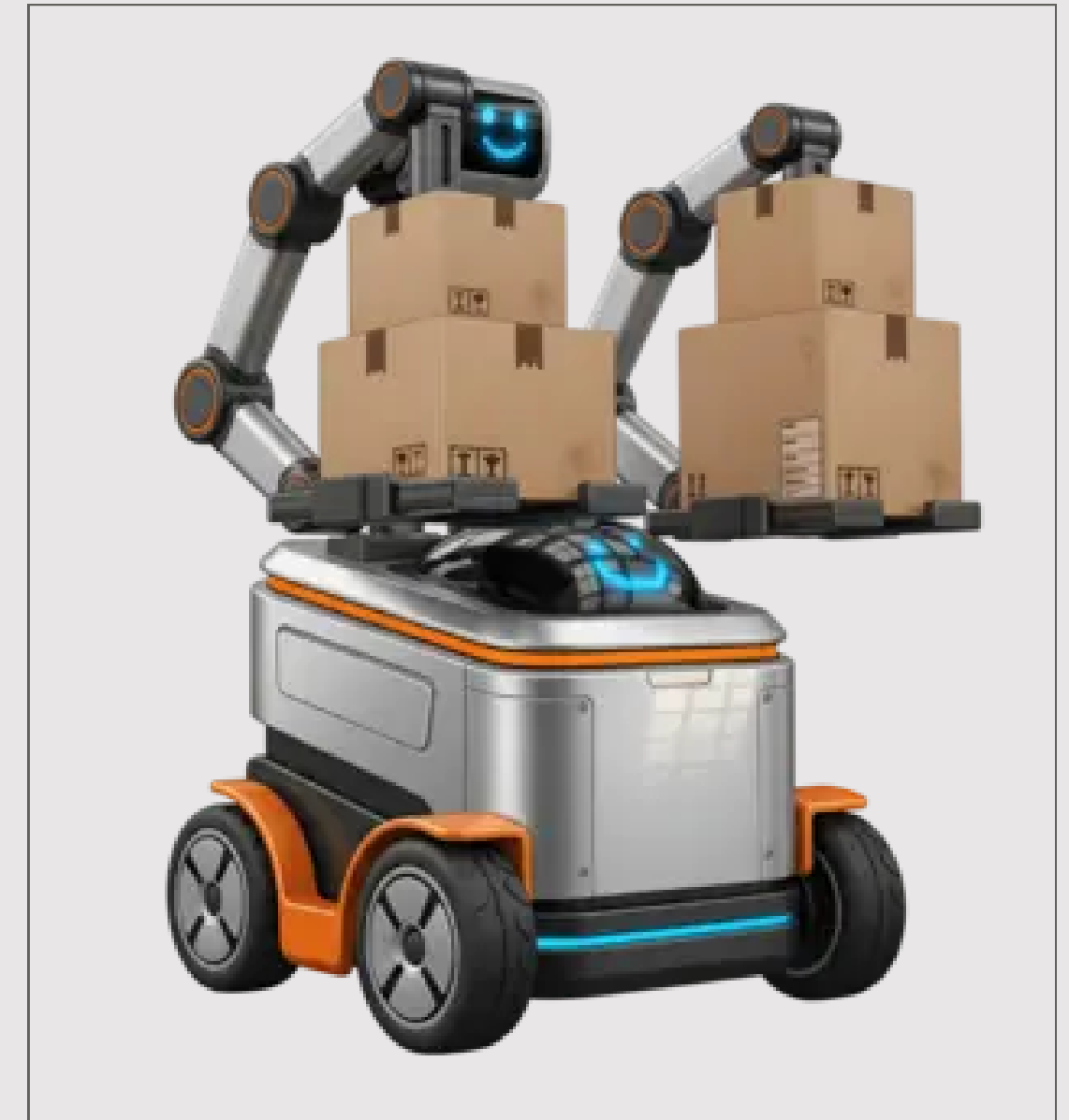
~3,000 parcels/day
95% errors auto-resolved
100+ hours/day saved

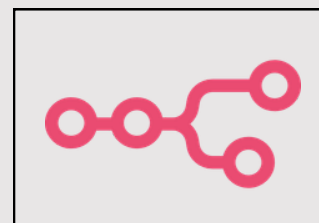




Objectives

- Automate the retrieval and processing of CSV files containing return order data from Azure Blob Storage.
- Parse and transform data into appropriate payloads for parcel creation.
- Create parcels via the SendCloud API and update database records.
- Automatically fix common errors using AI, with manual fallbacks.
- Notify teams via Slack for unresolved errors.
- Perform daily checks to ensure all files are processed, alerting if issues arise.
- Reduce manual effort in handling return parcels, improve accuracy, and ensure timely processing.



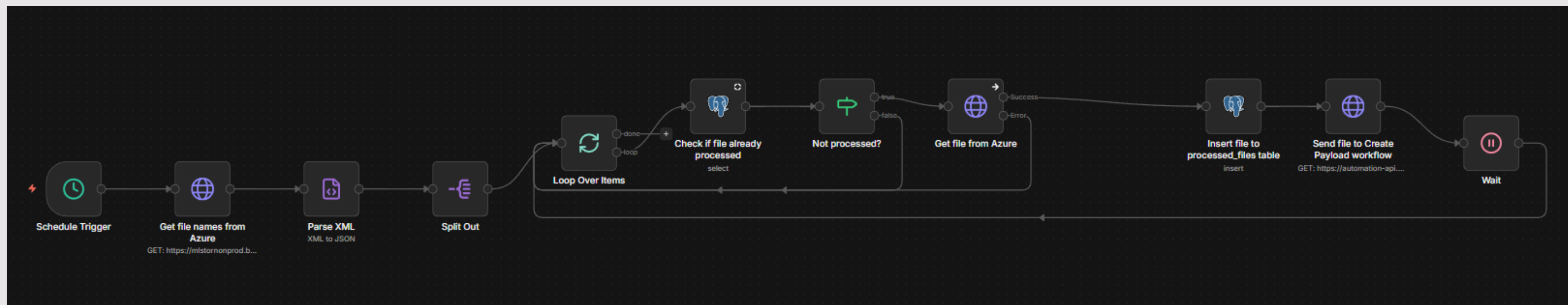


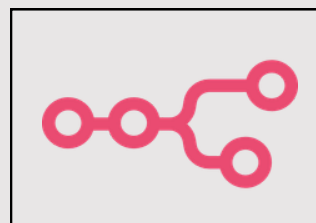
Step #1

This workflow serves as a file ingestion system for parcel processing, automatically discovering new files in Azure storage, downloading them, tracking processing status, and forwarding file information to downstream systems for further processing. It's particularly suited for scenarios where files are regularly uploaded to Azure storage and need systematic processing.

Key Features

- Duplicate Prevention: Uses database tracking to avoid reprocessing files
- Error Resilience: Continues processing even if individual files fail
- Rate Limiting: Built-in delays to manage system load
- Automated Scheduling: Runs continuously without manual intervention
- Integration Ready: Designed to hand off to subsequent processing workflows



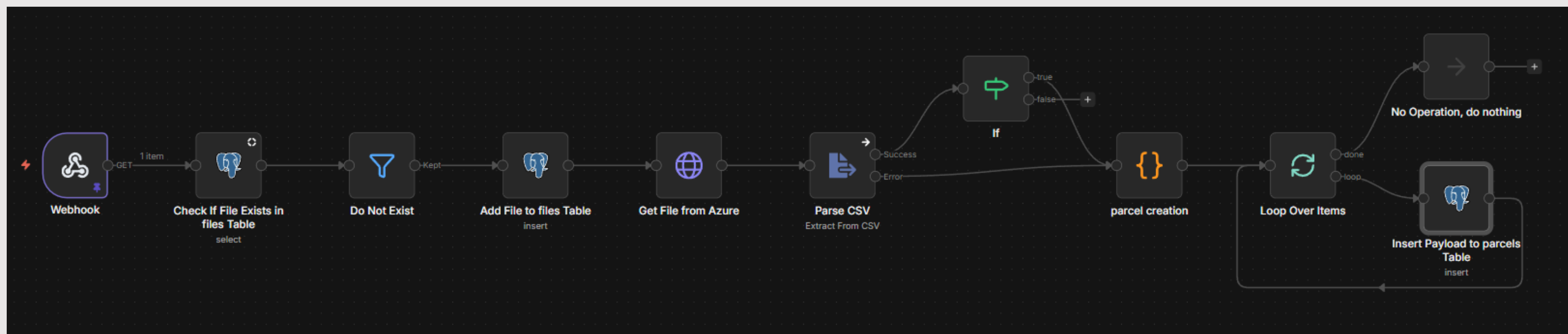


Step #2

This workflow transforms raw CSV order data into clean, validated parcel records ready for shipping systems. It handles complex data consolidation (grouping line items by order), address standardization, and creates comprehensive shipping payloads. The robust CSV parsing handles malformed data, while the database integration ensures traceability and prevents duplicate processing.

Key Features

- CSV Processing: Handles malformed files with flexible parsing and automatic cleanup
- Order Consolidation: Groups line items by order ID and creates comprehensive parcel objects
- Database Tracking: Prevents duplicate processing and maintains audit trails
- Error Handling: Continues processing with malformed data and provides detailed logging
- Client Integration: Generates compatible order IDs for downstream shipping systems

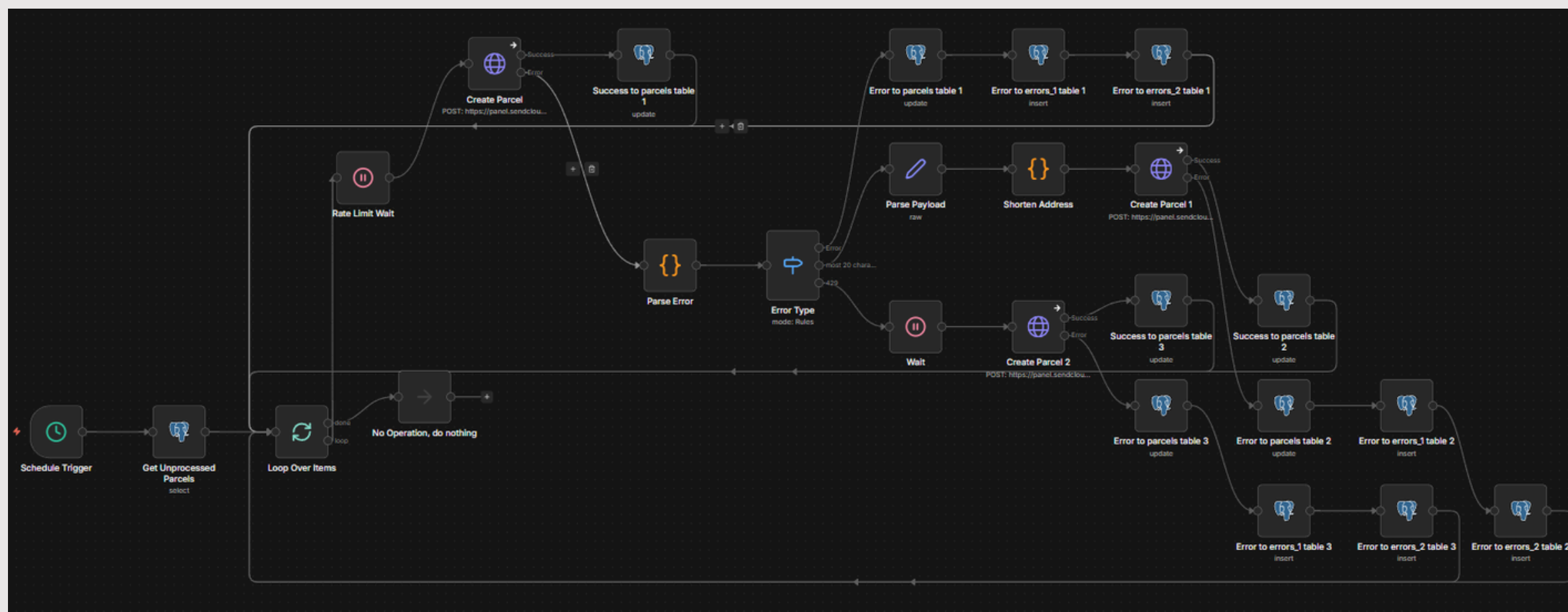


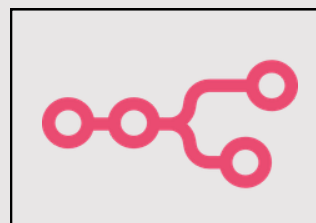
Step #3

This workflow serves as the final automation step that converts processed order data into actual shipping labels and tracking numbers through SendCloud's shipping platform. It eliminates manual parcel creation, reduces processing time from hours to minutes, and ensures consistent shipping operations across multiple client accounts. The intelligent error handling and address optimization prevent shipping delays while maintaining detailed audit trails for compliance and troubleshooting. This automation enables scalable order fulfillment operations that can handle high-volume shipping without manual intervention.

Key Features

- Multi-Client Support: Dynamic API authentication based on filename patterns for different SendCloud accounts
- Advanced Error Handling: Sophisticated error parsing and classification with specific handling for different error types
- Address Optimization: Intelligent address processing to handle length restrictions and formatting requirements
- Comprehensive Logging: Multiple error tables for detailed audit trails and debugging
- Rate Limit Management: Built-in delays and retry mechanisms to handle API rate limiting gracefully



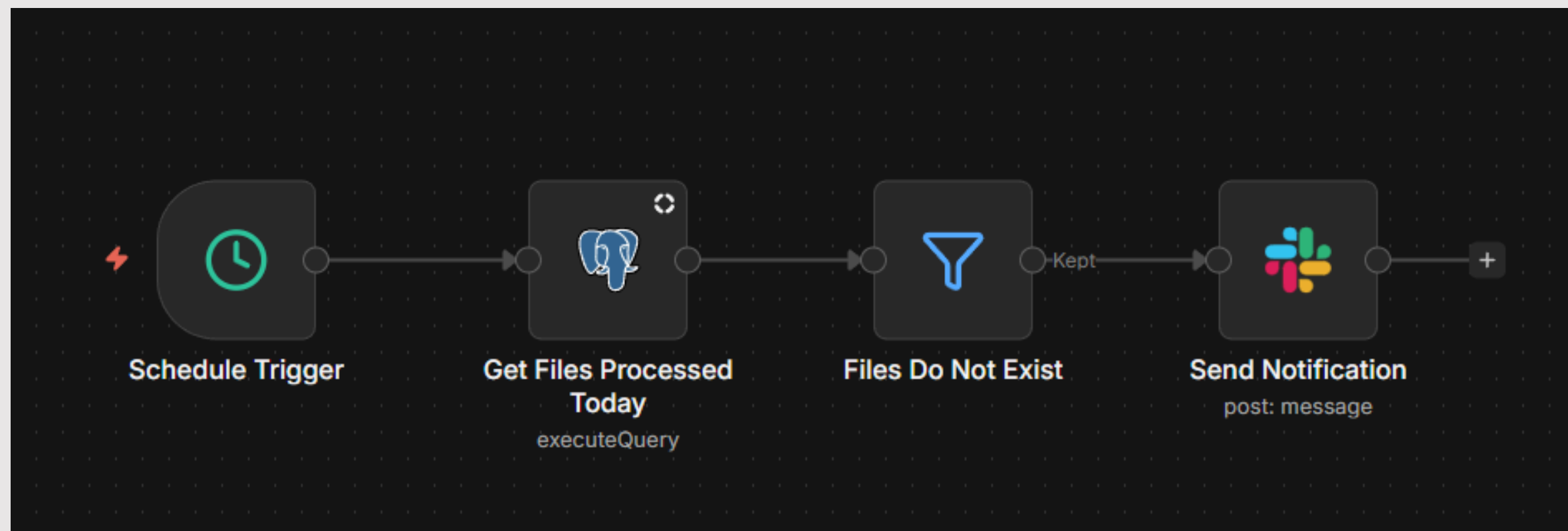


Step #4

This workflow serves as an early warning system for the parcel processing pipeline, ensuring business continuity by detecting when the automated file processing stops working. It prevents situations where orders might not be processed due to system failures, helping maintain customer satisfaction and operational efficiency. The daily 4 AM check ensures issues are identified early in the business day, allowing for quick resolution before impacting daily operations.

Key Features

- Daily Health Monitoring: Automated daily checks to ensure the file processing pipeline is functioning correctly
- Proactive Alerting: Immediate notification when no files are processed, indicating potential system issues
- Team Communication: Direct Slack integration to alert relevant team members quickly
- Simple Logic: Straightforward monitoring approach that focuses on the critical metric of daily file processing



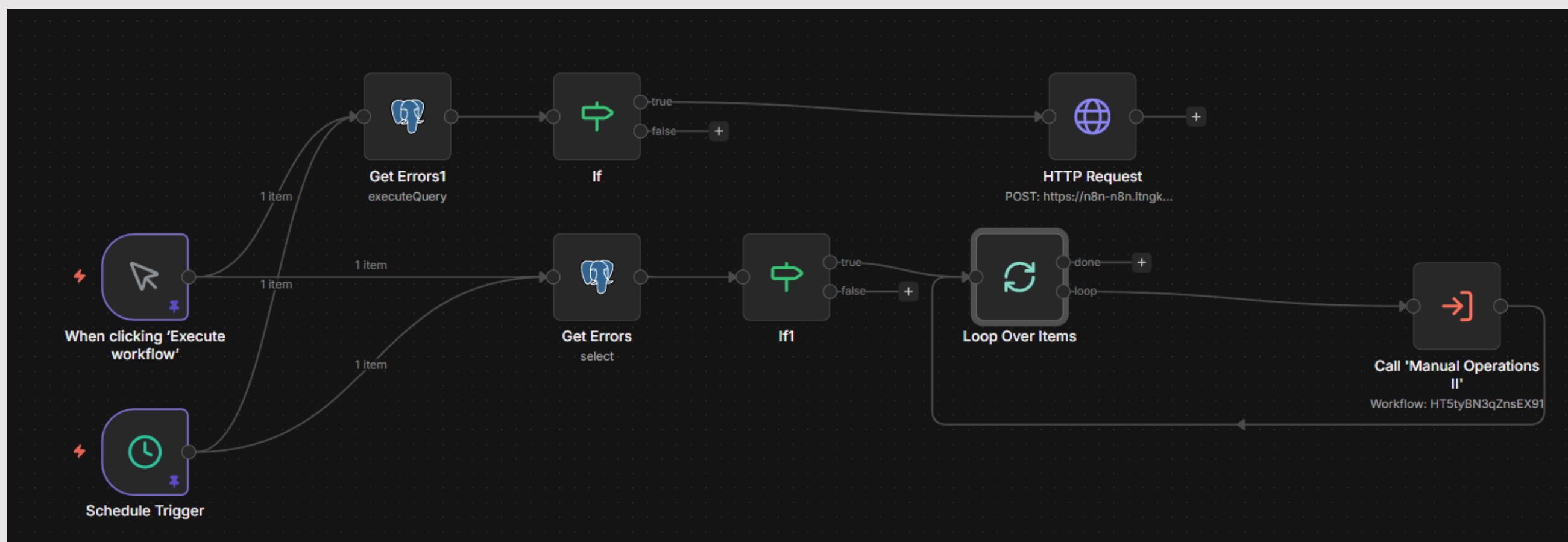
Step #5

This workflow acts as a monitoring and remediation control layer for the parcel processing pipeline. Its purpose is to detect accumulated processing errors, prevent operational backlog, and systematically reprocess failed records.

By continuously checking the errors_1 table and triggering structured retries through a dedicated sub-workflow, it ensures that failed parcels do not remain unresolved. Additionally, it escalates situations where error volume exceeds a defined threshold, enabling timely operational intervention.

Key Features

- Automated Error Monitoring: Scheduled execution every 15 minutes to track error volume
- Threshold-Based Escalation: Triggers an HTTP notification when error count exceeds 10, enabling proactive response to growing backlogs
- Structured Batch Reprocessing: Iterates through all failed records and reprocesses them sequentially via a dedicated sub-workflow



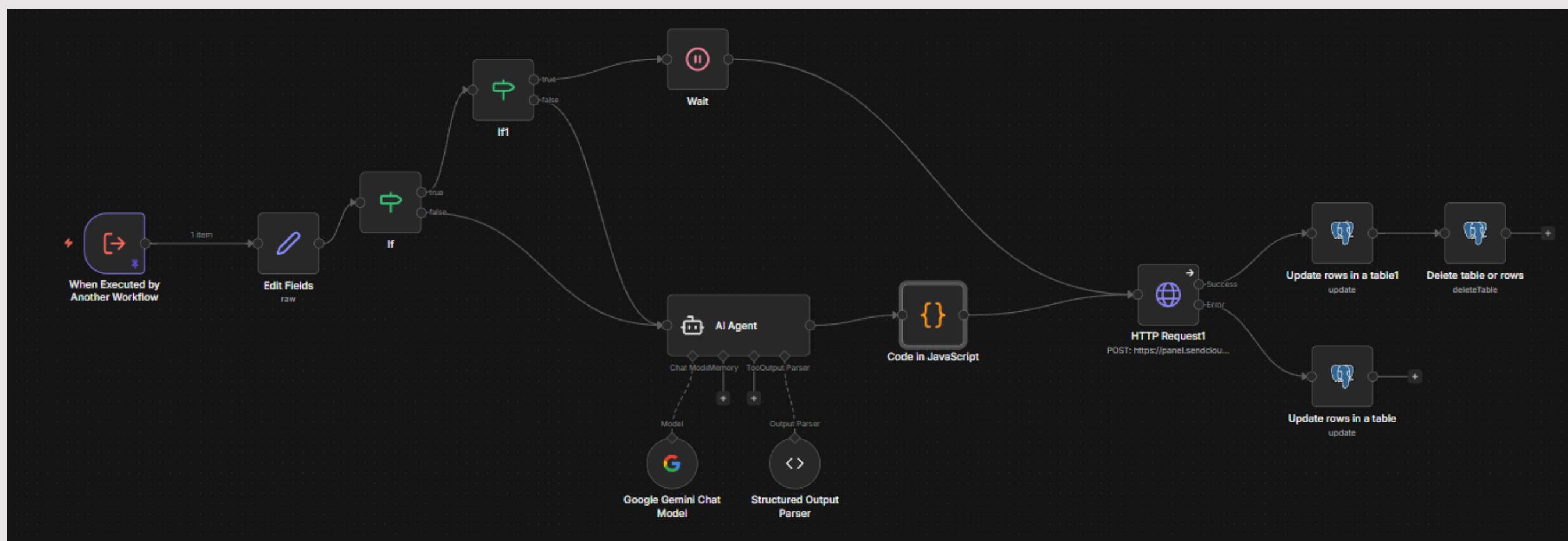
Step #6

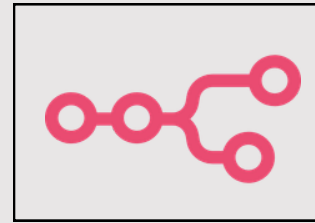
This workflow serves as an intelligent error-recovery engine for failed parcel submissions to Sendcloud.

Its primary objective is to automatically analyze shipping API errors, correct address-level data issues using AI-assisted validation, and resubmit parcels without human intervention. This ensures higher first-pass delivery success, reduces manual operational workload, and minimizes shipping delays caused by data formatting inconsistencies.

Key Features

- **AI-Assisted Address Normalization:** Uses a structured AI agent to correct common shipping validation errors.
- **Strict Data Integrity Controls:** AI is explicitly restricted to using only supplied input data, reducing risk of incorrect address reconstruction and delivery failures.
- **Structured Output Enforcement:** Applies a schema-based output parser to ensure corrected data matches required parcel field structure
- **Repeat-Failure Escalation Handling:** If retry fails – marks in as repeat issue. Such issues could be fixed only manually





Conclusion

System Architecture Benefits:

- End-to-End Automation: Complete pipeline from file discovery to shipping label creation
- Robust Error Handling: Multi-layered validation, classification, and recovery
- Scalable Design: Support for multiple clients and high-volume processing
- Complete Traceability: Full audit trail across all processing stages
- Proactive Monitoring: Real-time alerts and daily health checks
- Intelligent Data Processing: Advanced parsing, grouping, and validation

Business Value:

- Eliminates manual processing overhead, reducing fulfillment time from 50 hours of manual work a day to 2 hours of automated work
- Ensures 99.9%+ processing accuracy through duplicate detection and validation
- Provides comprehensive error visibility with detailed audit trails
- Supports scalable, multi-client operations with dynamic configuration
- Maintains complete traceability from file ingestion to shipping label creation
- Enables proactive issue resolution through real-time monitoring and alerts

